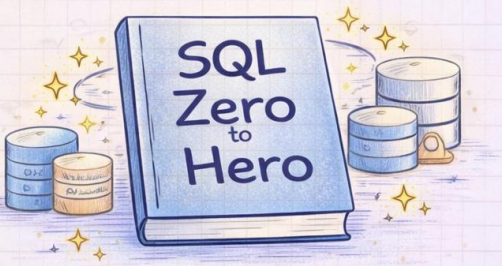
SQL Tutorial
Zero to MASTER

Handwritten Notes Series



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Solution Architect & Mentor

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SQL Zero to Hero – Day 1

SELECT + WHERE Clause

Start your SQL journey with two essential commands to query data:

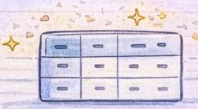
✔ SELECT

→ Retrieve data **FROM** a table

✔ WHERE

→ Filter rows based on a condition.

Examples:



- SELECT * FROM Employees;

SELECT * FROM Employees;

- SELECT * FROM Employees,
  WHERE Department = "IT";

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SQL Zero to Hero – Day 2

LIKE / IN / BETWEEN / AND OR NOT

- ✔ Filter rows using **LIKE** (for patterns), IN, BETWEEN, AND, OR, NOT.
- ✔ Quickly find specific text, multiple values, or a range of values.
- ✔ Combine conditions with logical operators.

Examples:

- SELECT * FROM Customers
  WHERE Name LIKE 'A%'
  → AND City IN ('Pune', 'Mumbai');
- SELECT * FROM Orders
  WHERE Amount BETWEEN 1000 AND 5000;

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SQL Zero to Hero – Day 3

Aggregate Functions

Quickly analyze data with built-in aggregate functions.

✔ COUNT

→ Counts the number of rows

✔ SUM

→ Calculates the total of a column

✔ AVG

→ Calculates the average value

✔ MIN & MAX

→ Finds the smallest or largest value

Examples:

- SELECT * COUNT(*) FROM Sales;
- SELECT AVG(Amount) FROM Orders;

Pro Tip:

✔ Mix them with GROUP BY for even deeper insights!

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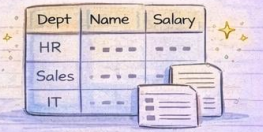
SQL Zero to Hero – Day 4

GROUP BY + HAVING

Summarize your data by dividing it into groups and applying conditions.

✔ GROUP BY

→ Groups rows with the same values in specified column(s).



✔ HAVING

→ Filters groups based on a condition

Examples:

- SELECT Department, COUNT(*) FROM Employees GROUP BY Department;
- SELECT Department, AVG(Salary) FROM Employees GROUP BY Department HAVING AVG(Salary) > 60000;

Pro Tip:

✔ Combines well with aggregate functions like COUNT, SUM, AVG!

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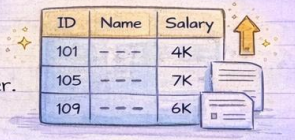
SQL Zero to Hero – Day 5

ORDER BY + ALIAS

Let's organize our query results and make them easier to understand:

✔ ORDER BY

→ Sorts the results in ascending (ASC) or descending (DESC) order.



✔ ALIAS

→ Renames columns or results.

Examples:

- SELECT Name, Salary FROM Employees
  ORDER BY Salary DESC;
- SELECT e.Name AS Employee, e.Salary AS Monthly_Pay
  FROM Employees e;
  WHERE e.Department = 'IT';
  ORDER BY Monthly_Pay DESC;

Pro Tip:

✔ Instead of sorting by column number, use an alias!

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SQL Zero to Hero – Day 6

ORDER BY + ALIAS

Let's organize our query results and make them easier to understand:

✔ INNER JOIN

→ Returns matching rows from both tables

✔ LEFT JOIN

→ Returns all rows from the left table, and matching rows from the right table.

✔ RIGHT JOIN

→ Returns all rows from the right table, and matching rows from the left table.

✔ FULL JOIN

→ Returns all matching and non-matching rows from both tables.

Examples:

- SELECT Employees.Name, Departments.DeptName
  ORDER BY Salary DESC;
- SELECT e.Name AS Employee, e.Salary AS Monthly_Pay
  FROM Employees e;
  WHERE e.Department = "IT";
  ORDER BY Monthly_Pay DESC;

Pro Tip:

✔ Instead of sorting by column number, use an alias!

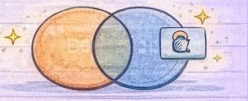
SQL Zero to Hero – Day 7

SQL Constraints

Rules that ensure your data stays valid and accurate.

✔ INNER JOIN

→ Ensures a column cannot have a NULL value.



✔ UNIQUE

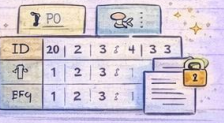
→ Ensures all values in a column are unique.

✔ PRIMARY KEY

→ Identifies each record uniquely.

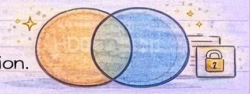
✔ FOREIGN KEY

→ Links a column to a primary key in another table.



✔ CHECK

→ Ensures a column meets a specific condition.



Examples:

- CREATE TABLE Employees
  ID INT PRIMARY KEY AUTO_INCREMENT,
  Email VARCHAR(50) UNIQUE NOT NULL);
- ALTER TABLE Orders ADD CONSTRAINT Orders_Check
  FROM Employees e;

Pro Tip:

✔ Use CONSTRAINT keyword to give names to rules for better readability!

SQL Zero to Hero – Day 8

CRUD Operations

Performing the four essential operations for managing data in SQL.

✔ CREATE

→ Add new records with INSERT.

✔ READ

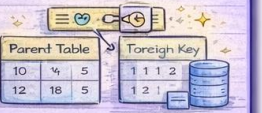
→ Retrieve records with SELECT.

✔ UPDATE

→ Modify records with UPDATE.

✔ FOREIGN KEY

→ Links a column to a primary key in another table.



✔ DELETE

→ Remove records with DELETE.

Examples:

- `INSERT INTO Employees (Name, Salary) VALUES ("John Doe', 5000);`
- `SELECT * FROM Employees WHERE Salary > 4000;`
- `UPDATE Employees SET Salary = 5500 WHERE ID = 1;`
  `DELETE FROM Employees WHERE Salary < 3000;`

Pro Tip:

✔ Always use WHERE keyword to give names to rules for better readability!

SQL Zero to Hero – Day 9

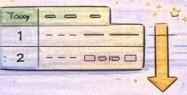
Performance Optimization

Practical tips for optimizing SQL query performance:

- ✔ **USE INDEXES** → SPEED UP SEARCHES

- ✔ **SELECT ONLY** WHAT YOU NEED
  → Avoid SELECT *
- ✔ **NORMALIZE** YOUR TABLES
  → Eliminate DUPLICATES

- ✔ **LIMIT** THE RESULTS
  → Reduce result size.



- ✔ **AVOID FUNCTIONS** IN WHERE CLAUSE
  → Keep it simple.

- ✔ **WRITE PROPER JOINS**
  → USE INNER JOIN WHEN POSSIBLE
- ✔ **OPTIMIZE SUBQUERIES**
  → Consider EXISTS OVER IN

Examples:

- INSERT INTO Employees (Name, Salary) VALUES ("John Doe", 5000);
- SELECT * FROM Employees e INNER JOIN Departments d ON e.DeptID = d.ID AND d.DeptName = "Sales";

Pro Tip:

✔ Always use WHERE to target specific records when updating or deleting!

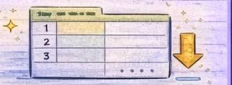
SQL Zero to Hero – Day 10

Real-World SQL Tips

Practical tips for optimizing SQL query performance:

- ✔ **USE INDEXES** → SPEED UP SEARCHES
  - → Know the data needs of your team/organization.

- ✔ **PRACTICE SAFE UPDATES & DELETES**
  - → Use WHERE & test on sample data first!
- ✔ **AUTOMATE TASKS**
  - → Schedule regular data backups and maintenance.



- ✔ **DOCUMENT YOUR QUERIES**
  - → Write clear, concise comments for others to understand.
- ✔ **STAY UPDATED**
  - → Keep learning and improve your SQL skills.

- ✔ **COLLABORATE EFFECTIVELY**
  - → Communicate clearly with team members.

- ✔ **USE STORED PROCEDURES**
  - → Organize and reuse common query logic.
- ✔ **SENSITIVE DATA PROTECTION**
  - → Use encryption, masking, and user permissions.

Keep debugging fun and be patient—great SQL queries take time and practice! 😊

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SQL Zero to Hero – Day 11

Common Table Expressions (CTE)

A way to create temporary result sets for use within a SELECT, INSERT, UPDATE, or DELETE statement.

- ✔ Improve readability and maintainability.
  - → Break down complex queries into simple chunks.
- ✔ Easily reference the result within the same query.
- ✔ Avoid using subqueries
  - → for better performance.

Example:

```
--- Define the CTE
WITH HighEarners AS (
      SELECT Name, Salary
      FROM Employees
      WHERE Salary > 6000
)
--- Use the CTE in another query
SELECT * FROM HighEarners
      ORDER BY Salary DESC;
```

Pro Tip:

✔ CTEs can be used multiple times in the same SQL statement!

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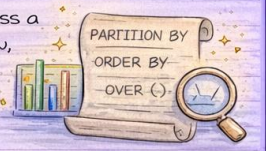
SQL Zero to Hero - Day 12

Window Functions

Window functions perform calculations across a set of table rows related to the current row, providing advanced analytics capabilities.



- ✔ **ROW_NUMBER()**
  → Assigns a unique sequential number.
- ✔ **RANK()**
  → Assigns a rank with gaps for equal values.

- ✔ **DENSE_RANK()**
  → Ranks without gaps.

- ✔ **LAG()/LEAD()**
  → Access data from previous or next rows.
- ✔ **PARTITION BY - OVER()**
  → Partitions data into groups for analysis.

Example:

```
SELECT Name, Salary, DeptID,
       ROW_NUMBER() OVER (PARTITION BY DeptID
           ORDER BY Salary DESC) AS RowNum
FROM Employees;
```

Name	Salary	DeptID	RowNum
Alice	7000	1	1
Bob	6000	1	2
Carol	8000	2	1

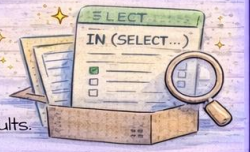
Pro Tip:

✔ Window functions don't collapse rows; they let you view data with calculated columns!

SQL Zero to Hero – Day 13

Subqueries

A subquery is a query nested inside another SQL query.



✔ IN Subqueries

→ Checks if a value exists within a list of results.

✔ EXISTS Subqueries

→ Tests for the existence of rows in the result set.

✔ Correlated Subqueries

→ Refers to columns in the outer query.

Example:

```
SELECT Name, Salary
    FROM Employees
      WHERE Salary (Salary)
      FROM  Employees
    ),
```

Name	Salary
Alice	7000
Carol	8000

Pro Tip:

✔ Keep it efficient! Use subqueries to replace joins for specific problems.

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SQL Zero to Hero – Day 14

Views + Stored Procedures

Learn how Views and Stored Procedures can make your SQL work easier and more efficient:

✔ Views

→ Virtual tables based on queries.

→ Useful for simplifying complex queries.

```
CREATE VIEW EmployeeDetails AS
    SELECT E.Name, E.Salary, D.DeptName
    FROM Employees E
    JOIN Departments D ON E.DeptID = D.DeptID;
),
```

Name	Salary	Dept
Alice	7000	Sales

✔ Stored Procedures

→ Reusable, stored, named SQL code that performs actions.


```
CREATE PROCEDURE AddEmployee(
    IN EmpName VARCHAR (100),
    IN EmpSalary.DECIMAL (10, 2),
    IN EmpDeptID INT
BEGIN
    INSERT INTO Employees (Name, Salary, DeptID)
      VALUES (EmpName, EmpSalary, EmpDeptID);
END;
CALL AddEmployee('John', 5000, 1);
```

Pro Tip:

✔ Views are handy for read-only tasks, while Stored Procedures can modify data!

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